Exponential and Logarithmic Equations

ANSWER KEY

Exponential equations

1 Solve $3^{x+1} = 81$ exactly.
$3^{x+1} = 3^4$, so $x + 1 = 4$ and $x = 3$.

2 Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.
$2x = \log_5 17 = \frac{\ln 17}{\ln 5}$, so $x = \frac{\ln 17}{2\ln 5} \approx 0.880$.

3 Solve $e^{2x} - 5e^x + 6 = 0$ exactly.
Let $u = e^x$: $u^2 - 5u + 6 = (u-2)(u-3) = 0$, so $e^x = 2$ or $e^x = 3$: $x = \ln 2$ or $x = \ln 3$.

Logarithmic equations

4 Solve $\ln(2x-3) = 1$. Give an exact answer and a decimal to three places.
$2x - 3 = e$, so $x = \frac{e+3}{2} \approx 2.859$.

5 Solve $\log_2 x + \log_2 (x-6) = 4$, checking for extraneous solutions.
$\log_2 (x(x-6)) = 4$, so $x^2 - 6x - 16 = 0$, giving $(x-8)(x+2) = 0$. Reject $x = -2$ (logarithm of a negative number): $x = 8$.

6 Solve $\log(x+3) - \log(x-1) = 1$, checking for extraneous solutions.
$\frac{x+3}{x-1} = 10$, so $x + 3 = 10x - 10$ and $x = \frac{13}{9}$. Check: $x > 1$, valid.

7 The pH of a solution is $\mathrm{pH} = -\log[\mathrm{H}^+]$. Find the hydrogen-ion concentration $[\mathrm{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.
$[\mathrm{H}^+] = 10^{-4.6} \approx 2.5 \times 10^{-5}$ mol/L.

More exponential equations

8 Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.
Take logs: $x\ln 2 = (x-1)\ln 3$, so $x(\ln 2 - \ln 3) = -\ln 3$, giving $x = \frac{\ln 3}{\ln 3 - \ln 2} = \frac{\ln 3}{\ln(3/2)} \approx \frac{1.0986}{0.4055} \approx 2.710$.

9 Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.
$4^x = (2^x)^2 = u^2$: $u^2 - 3u - 4 = 0$, so $(u-4)(u+1) = 0$. $u = 4$ (rejecting $u = -1$, since $2^x > 0$): $2^x = 4$, so $x = 2$.

Systems and applications

10 A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

$2000 = 500e^{6k}$: $e^{6k} = 4$, so $k = \frac{\ln 4}{6} \approx 0.2310$. For $N = 10000$: $10000 = 500e^{0.2310t}$, so $e^{0.2310t} = 20$, $t = \frac{\ln 20}{0.2310} \approx 12.97 \approx 13.0$ hours.

11 Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

Multiply by $u = e^x$: $u^2 - 3u + 1 = 0$, so $u = \frac{3 \pm \sqrt{5}}{2}$. Both roots are positive, giving $x = \ln\left(\frac{3+\sqrt{5}}{2}\right)$ or $x = \ln\left(\frac{3-\sqrt{5}}{2}\right)$.

Combined logarithmic and exponential practice

12 Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous solutions.

$\log_3((x+2)(x-2)) = 2$: $x^2 - 4 = 9$, so $x^2 = 13$, $x = \pm\sqrt{13}$. Reject $x = -\sqrt{13}$ (makes $x - 2 < 0$): $x = \sqrt{13}$.

13 Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

$3^{2x} = 3^{3(x-1)}$: $2x = 3x - 3$, so $x = 3$.

Equations requiring both exponential and logarithmic techniques

14 Solve $\log_2(x) = 3 - \log_2(x-2)$, checking for extraneous solutions.

$\log_2 x + \log_2(x-2) = 3 \Rightarrow \log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Reject $x = -2$ (negative argument): $x = 4$.

15 Solve $e^{\ln x + \ln 2} = 10$ for x, simplifying the exponent first.

$e^{\ln x + \ln 2} = e^{\ln(2x)} = 2x$ (since $e^{\ln u} = u$). So $2x = 10 \Rightarrow x = 5$.

Verifying solutions numerically

16 For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.

$5^{2(0.880)} = 5^{1.76} \approx 17.02$, matching the right side 17 (small rounding difference due to the decimal approximation of x), confirming the solution.

Equations with different bases on each side

17 Solve $7^x = 3^{x+2}$, giving an exact expression and a decimal to three places.

Take logs: $x\ln 7 = (x+2)\ln 3$: $x(\ln 7 - \ln 3) = 2\ln 3$, so $x = \frac{2\ln 3}{\ln 7 - \ln 3} = \frac{2\ln 3}{\ln(7/3)} \approx \frac{2.1972}{0.8473} \approx 2.593$.

18 Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

$2x - 1 = \log 500$, so $x = \frac{1 + \log 500}{2} \approx \frac{1 + 2.6990}{2} \approx 1.850$.

A modeling problem requiring both equation types

19 A population model is $P(t) = 800e^{0.05t}$. Find the time t when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

For $P = 2000$: $2000 = 800e^{0.05t} \Rightarrow 2.5 = e^{0.05t} \Rightarrow t = \frac{\ln 2.5}{0.05} \approx 18.33$. For $P = 1500$: $1500 = 800e^{0.05t} \Rightarrow 1.875 = e^{0.05t} \Rightarrow t = \frac{\ln 1.875}{0.05} \approx 12.58$.

Checking solutions by substitution

20 Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2 (x - 6) = 4$ by substituting back into the original equation.

$\log_2 (8) + \log_2 (8 - 6) = \log_2 (8) + \log_2 (2) = 3 + 1 = 4$ ✓, confirming $x = 8$ satisfies the original equation.

21 Solve $\ln(x) + \ln(x - 3) = \ln(10)$, checking for extraneous solutions.

$\ln(x(x - 3)) = \ln 10 \Rightarrow x^2 - 3x = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0$. Reject $x = -2$ (negative argument): $x = 5$.

22 A radioactive sample decays according to $A(t) = A_0 e^{-0.08t}$. If the initial amount is 500 g, find the time t (to two decimal places) at which only 50 g remains, and separately find the time at which 90% of the original sample has decayed away.

For $A = 50$: $50 = 500e^{-0.08t} \Rightarrow 0.1 = e^{-0.08t} \Rightarrow t = \frac{\ln 0.1}{-0.08} \approx 28.79$. For 90% decayed, 10% remains ($A = 50$ g as well, since 10% of 500 is 50): this is the same equation, so $t \approx 28.79$ — confirming both descriptions refer to the same instant.

23 Solve $4^x = 8^{x-1}$ exactly, writing both sides with base 2.

$2^{2x} = 2^{3(x-1)}$: $2x = 3x - 3$, so $x = 3$.

24 Solve $\log_5 (x + 4) = 2$ for x.

$x + 4 = 5^2 = 25$, so $x = 21$.